

Technical Supplement to Source-Attributed DLMP Settlement with Price-Anticipating Load Response

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Abstract

Under strong convexity and effective-space linear independence of the active constraint gradients, the smooth penalty minimizers and network multiplier estimates recover the centralized hard-constrained LinDistFlow optimum and its KKT multipliers in the iterated limit: first $\varepsilon \downarrow 0$ at fixed penalty scale, then $\kappa \rightarrow \infty$. This supplement proves the corresponding voltage- and congestion-price limits. Eliminating fixed load coordinates gives the nondegenerate decision space on which the regularity condition is imposed. A separate analysis derives the source-attribution feedback Jacobian, continuous path moments, and curvature bounds used to assess strong monotonicity of the responsibility-settlement VI. The OPF limiting result and the QNE uniqueness criterion concern different solution concepts and retain their respective assumptions.

Sections S1–S4 establish and interpret the centralized OPF limiting result in Theorem S1. Section S5 provides the technical derivations for the main manuscript’s QNE monotonicity analysis.

S1 Effective-space hard-constrained problem

S1.1 Elimination of rigid coordinates

Let the flexible coordinates be collected in $\mathbf{y} \in \mathcal{X} \subset \mathbb{R}^d$, where

$$\mathcal{X} := \prod_{h=1}^d [\underline{y}_h, \bar{y}_h], \quad \underline{y}_h < \bar{y}_h. \quad (\text{S1})$$

Here $(\mathbf{y}, \mathcal{X}, d)$ corresponds to the manuscript’s $(\mathbf{y}_a, \mathcal{Y}_a, n_a)$. The embedding matrix G_a and fixed-load vector $\mathbf{p}^{\text{fix}}$ retain their manuscript definitions. Fixed-load consumption, independent of the settlement baseline, enters the full non-root-bus power vector through

$$\mathbf{p}(\mathbf{y}) = \mathbf{p}^{\text{fix}} + G_a \mathbf{y}. \quad (\text{S2})$$

Each retained interval has positive width, so the constraint qualification below is imposed directly on the free coordinates.

Define the reduced centralized objective

$$F(\mathbf{y}) := \mathcal{S}(\mathbf{p}(\mathbf{y})), \quad (\text{S3})$$

where $\mathcal{S}$ is the centralized welfare objective in the manuscript. The scalar F in this supplement is distinct from the manuscript’s vector VI operators F_p and F_y . Fixed-load utility constants do not affect the minimizers or derivatives. Under the main-study assumptions,

$$\nabla^2 F(\mathbf{y}) = D_{\text{util},a} + \frac{2\pi^E}{v_0} G_a^\top H^L G_a \succ 0, \quad (\text{S4})$$

because $D_{\text{util},a} = \text{diag}(\beta_{u,h}) \succ 0$, $\pi^E \geq 0$, and $H^L \succeq 0$. Hence F is continuously differentiable and strongly convex on $\mathcal{X}$.

For the bus-specific lower-voltage limits $\underline{v}_m$ and line-specific forward-flow limits $\bar{f}_\ell$ of the main studies, set

$$g_m^V(\mathbf{y}) := \underline{v}_m - v_m(\mathbf{p}(\mathbf{y})), \quad (\text{S5})$$

$$g_\ell^C(\mathbf{y}) := f_\ell(\mathbf{p}(\mathbf{y})) - \bar{f}_\ell. \quad (\text{S6})$$

Stack all network inequalities as $g_i(\mathbf{y}) \leq 0$, $i = 1, \dots, r$. These functions are affine. Let

$$G := [\nabla g_1 \ \cdots \ \nabla g_r] \in \mathbb{R}^{d \times r}. \quad (\text{S7})$$

The hard-constrained effective-space problem is

$$\min_{\mathbf{y} \in \mathcal{X}} F(\mathbf{y}) \quad \text{subject to} \quad g(\mathbf{y}) \leq 0. \quad (\text{S8})$$

This is Model B in the manuscript; its smooth outer-penalty counterpart below is Model C. Both retain the same physical load bounds and fixed consumption.

S1.2 KKT multipliers and scarcity-price mapping

Assume that problem (S8) is feasible, and denote its unique minimizer by $\mathbf{y}^*$. The uniqueness follows from Eq. (S4). Let

$$\mathcal{A}_N^* := \{i : g_i(\mathbf{y}^*) = 0\} \quad (\text{S9})$$

be the active network-constraint set. Let B^* collect one outward normal for every active bound of the nondegenerate box $\mathcal{X}$ at $\mathbf{y}^*$. We impose the effective-space LICQ

$$[B^* \ G_{\mathcal{A}_N^*}] \quad \text{has full column rank}, \quad (\text{S10})$$

where $G_{\mathcal{A}_N^*}$ contains the columns of G indexed by $\mathcal{A}_N^*$. Unlike a full-space LICQ imposed on two inequalities for a zero-width interval, Eq. (S10) is compatible with rigid loads because those coordinates have already been eliminated.

The hard KKT system is

$$0 \in \nabla F(\mathbf{y}^*) + G\boldsymbol{\omega}^h + N_{\mathcal{X}}(\mathbf{y}^*), \quad (\text{S11})$$

$$g(\mathbf{y}^*) \leq 0, \quad \boldsymbol{\omega}^h \geq 0, \quad (\text{S12})$$

$$\omega_i^h g_i(\mathbf{y}^*) = 0, \quad i = 1, \dots, r. \quad (\text{S13})$$

Here $N_{\mathcal{X}}$ is the convex-analysis normal cone. Condition (S10) makes the complete KKT multiplier vector, and hence the network multiplier vector $\boldsymbol{\omega}^h$, unique.

Partition $\boldsymbol{\omega}^h$ into the low-voltage multipliers $\boldsymbol{\lambda}^h$ and the forward-flow multipliers $\boldsymbol{\nu}^h$. Since

$$\frac{\partial g_m^V}{\partial p_k} = A_{mk}^V, \quad \frac{\partial g_\ell^C}{\partial p_k} = A_{\ell k}^C, \quad (\text{S14})$$

the corresponding full-power marginal terms are

$$\pi_k^{V,*} = \sum_m A_{mk}^V \lambda_m^h, \quad \pi_k^{C,*} = \sum_\ell A_{\ell k}^C \nu_\ell^h. \quad (\text{S15})$$

This establishes the sign and sensitivity structure used in the hard DLMP.

S1.3 Numerical regularity check for the centralized cases

For the three centralized hard-OPF cases in Numerical Study I, the matrix in Eq. (S10) was reconstructed from the saved optimal profiles and current model inputs, without resolving the OPF. Coordinates with interval width at most 10^{-12} were eliminated. Active sets were identified using absolute tolerances 10^{-7} , 10^{-8} , and 10^{-9} ; each gradient column was normalized to unit Euclidean norm before applying the numerical rank threshold 10^{-9} . All $3 \times 24 \times 3 = 216$ case-hour-tolerance checks gave full column rank. The smallest singular values of the column-normalized matrices, over the 24 hours and all three active-set tolerances, were 1.000000, 0.650115, and 0.370054 for Cases 1, 2, and 3, respectively. The script and input-hash records accompany this supplement in `validation/`. These floating-point diagnostics support the applicability of the regularity assumption to the tested profiles; they do not replace the exact LICQ assumption in the propositions below.

S2 Quadratic and smooth outer-penalty problems

S2.1 Positive-part smoothing

For $z \in \mathbb{R}$, let

$$q(z) := \frac{1}{2}[z]_+^2, \quad s_\varepsilon(z) := \varepsilon \log(1 + e^{z/\varepsilon}), \quad \varphi_\varepsilon(z) := \int_{-\infty}^z s_\varepsilon(t) dt. \quad (\text{S16})$$

The function q is continuously differentiable with $q'(z) = [z]_+$, whereas $\varphi_\varepsilon \in C^\infty$ and $\varphi'_\varepsilon = s_\varepsilon$. The standard Softplus bounds are

$$0 \leq s_\varepsilon(z) - [z]_+ \leq \varepsilon \log 2, \quad z \in \mathbb{R}. \quad (\text{S17})$$

Moreover,

$$0 \leq \varphi_\varepsilon(z) - q(z) \leq \frac{\pi^2}{6}\varepsilon^2, \quad z \in \mathbb{R}. \quad (\text{S18})$$

Indeed, the derivative of the difference in Eq. (S18) is nonnegative, and its total increase over $\mathbb{R}$ is

$$2\varepsilon^2 \int_0^\infty \log(1 + e^{-u}) du = \frac{\pi^2}{6}\varepsilon^2. \quad (\text{S19})$$

These bounds also show why smoothing errors in the multiplier coefficients are scaled by the penalty parameter.

S2.2 Penalized minimizers and multiplier estimates

For this proof, parameterize the manuscript's penalties at fixed positive ratios as $\kappa_V = \rho_V \kappa$ and $\kappa_C = \rho_C \kappa$. Each constraint weight ρ_i is either ρ_V or ρ_C . The quadratic outer-penalty problem is

$$\mathbf{y}^{(\kappa)} := \operatorname{argmin}_{\mathbf{y} \in \mathcal{X}} \left\{ F(\mathbf{y}) + \kappa \sum_{i=1}^r \rho_i q(g_i(\mathbf{y})) \right\}. \quad (\text{S20})$$

Its multiplier estimates are

$$\omega_i^{(\kappa)} := \kappa \rho_i [g_i(\mathbf{y}^{(\kappa)})]_+. \quad (\text{S21})$$

The smooth outer-penalty problem is

$$\mathbf{y}^{(\kappa, \varepsilon)} := \operatorname{argmin}_{\mathbf{y} \in \mathcal{X}} \left\{ F(\mathbf{y}) + \kappa \sum_{i=1}^r \rho_i \varphi_\varepsilon(g_i(\mathbf{y})) \right\}, \quad (\text{S22})$$

with smooth multiplier estimates

$$\omega_i^{(\kappa, \varepsilon)} := \kappa \rho_i s_\varepsilon(g_i(\mathbf{y}^{(\kappa, \varepsilon)})). \quad (\text{S23})$$

Strong convexity of F , convexity of q and φ_ε , and affinity of g_i make both penalized minimizers unique. Write P_κ and $P_{\kappa, \varepsilon}$ for the respective objectives in Eqs. (S20) and (S22). Their stationarity systems are

$$0 \in \nabla F(\mathbf{y}^{(\kappa)}) + G\boldsymbol{\omega}^{(\kappa)} + N_{\mathcal{X}}(\mathbf{y}^{(\kappa)}), \quad (\text{S24})$$

$$0 \in \nabla F(\mathbf{y}^{(\kappa, \varepsilon)}) + G\boldsymbol{\omega}^{(\kappa, \varepsilon)} + N_{\mathcal{X}}(\mathbf{y}^{(\kappa, \varepsilon)}). \quad (\text{S25})$$

For voltage and congestion constraints, Eq. (S23) is exactly

$$\mu_m^{(\kappa, \varepsilon)} = \kappa_V s_\varepsilon(v_m - v_m(\mathbf{p}(\mathbf{y}^{(\kappa, \varepsilon)}))), \quad \nu_\ell^{(\kappa, \varepsilon)} = \kappa_C s_\varepsilon(f_\ell(\mathbf{p}(\mathbf{y}^{(\kappa, \varepsilon)})) - \bar{f}_\ell). \quad (\text{S26})$$

S2.3 Implementation and objective normalization

The numerical implementation evaluates the shifted primitive

$$\tilde{\varphi}_\varepsilon(z) := \int_0^z s_\varepsilon(t) dt = \varphi_\varepsilon(z) - \frac{\pi^2}{12} \varepsilon^2. \quad (\text{S27})$$

Thus its penalized objective $\tilde{P}_{\kappa, \varepsilon}$ and the normalization used in this supplement satisfy

$$P_{\kappa, \varepsilon}(\mathbf{y}) = \tilde{P}_{\kappa, \varepsilon}(\mathbf{y}) + \frac{\pi^2}{12} \kappa \varepsilon^2 \sum_{i=1}^r \rho_i. \quad (\text{S28})$$

This additive constant leaves minimizers, gradients, multiplier estimates, and fixed-parameter candidate rankings unchanged. For n_V voltage rows and n_C congestion rows, this constant is

$$\frac{\pi^2 \varepsilon^2}{12} (n_V \kappa_V + n_C \kappa_C).$$

Absolute objective values from different models or parameter settings must therefore be placed on a common normalization before comparison. The nonnegative potential bound in Eq. (S18) and the convergence proofs below use φ_ε , as defined in Eq. (S16).

S3 Convergence results

S3.1 Removal of the smoothing error

Proposition S1 (Fixed- κ smoothing limit). *For every fixed $\kappa > 0$,*

$$\mathbf{y}^{(\kappa, \varepsilon)} \longrightarrow \mathbf{y}^{(\kappa)}, \quad \boldsymbol{\omega}^{(\kappa, \varepsilon)} \longrightarrow \boldsymbol{\omega}^{(\kappa)} \quad \text{as } \varepsilon \downarrow 0. \quad (\text{S29})$$

Proof. By Eq. (S18), the two objectives in Eqs. (S20) and (S22) satisfy the uniform estimate

$$0 \leq P_{\kappa, \varepsilon}(\mathbf{y}) - P_\kappa(\mathbf{y}) \leq \frac{\pi^2}{6} \kappa \varepsilon^2 \sum_{i=1}^r \rho_i, \quad \mathbf{y} \in \mathcal{X}. \quad (\text{S30})$$

Thus $P_{\kappa, \varepsilon} \rightarrow P_\kappa$ uniformly on the compact set $\mathcal{X}$. Every accumulation point of $\mathbf{y}^{(\kappa, \varepsilon)}$ minimizes P_κ . Since this minimizer is unique, the entire sequence converges to $\mathbf{y}^{(\kappa)}$.

For each i , Eq. (S17) and the 1-Lipschitz property of $[\cdot]_+$ give

$$\begin{aligned} \left| \omega_i^{(\kappa, \varepsilon)} - \omega_i^{(\kappa)} \right| &\leq \kappa \rho_i \varepsilon \log 2 \\ &\quad + \kappa \rho_i \left| g_i(\mathbf{y}^{(\kappa, \varepsilon)}) - g_i(\mathbf{y}^{(\kappa)}) \right|. \end{aligned} \quad (\text{S31})$$

Affinity of g_i and convergence of the minimizers make the right-hand side tend to zero. $\square$

S3.2 Quadratic-penalty limit

Proposition S2 (Quadratic outer-penalty consistency). *Under the assumptions in Section S1, including the effective-space LICQ in Eq. (S10),*

$$\mathbf{y}^{(\kappa)} \longrightarrow \mathbf{y}^*, \quad \boldsymbol{\omega}^{(\kappa)} \longrightarrow \boldsymbol{\omega}^h \quad \text{as } \kappa \rightarrow \infty. \quad (\text{S32})$$

Proof. Because $\mathbf{y}^*$ is feasible for the hard problem, optimality of $\mathbf{y}^{(\kappa)}$ yields

$$F(\mathbf{y}^{(\kappa)}) + \frac{\kappa}{2} \sum_{i=1}^r \rho_i [g_i(\mathbf{y}^{(\kappa)})]_+^2 \leq F(\mathbf{y}^*). \quad (\text{S33})$$

Let $F_{\min} := \min_{\mathbf{y} \in \mathcal{X}} F(\mathbf{y})$ and $\rho_{\min} := \min_i \rho_i > 0$. Compactness of $\mathcal{X}$ and continuity of F ensure that $F_{\min}$ is finite. Equation (S33) implies

$$\sum_{i=1}^r [g_i(\mathbf{y}^{(\kappa)})]_+^2 \leq \frac{2(F(\mathbf{y}^*) - F_{\min})}{\kappa \rho_{\min}} \longrightarrow 0. \quad (\text{S34})$$

Every accumulation point of $\mathbf{y}^{(\kappa)}$ is therefore hard-feasible. Also, Eq. (S33) gives $F(\mathbf{y}^{(\kappa)}) \leq F(\mathbf{y}^*)$. Hence every accumulation point is a global minimizer of the hard problem. Its uniqueness implies $\mathbf{y}^{(\kappa)} \rightarrow \mathbf{y}^*$.

It remains to prove multiplier convergence. If $i \notin \mathcal{A}_N^*$, then $g_i(\mathbf{y}^*) < 0$. Convergence of the primal sequence implies $g_i(\mathbf{y}^{(\kappa)}) < 0$ and consequently $\omega_i^{(\kappa)} = 0$ for all sufficiently large κ .

Let Z have orthonormal columns spanning the tangent subspace of the active box face at $\mathbf{y}^*$:

$$\text{range}(Z) = \ker((B^*)^\top). \quad (\text{S35})$$

Any bound inactive at $\mathbf{y}^*$ remains inactive at $\mathbf{y}^{(\kappa)}$ for sufficiently large κ . On a coordinate active at $\mathbf{y}^*$, the penalty minimizer is either interior or on the same bound once it is sufficiently close to $\mathbf{y}^*$. Therefore, for every sufficiently large κ and every $\mathbf{n}^{(\kappa)} \in N_{\mathcal{X}}(\mathbf{y}^{(\kappa)})$ appearing in Eq. (S24),

$$\mathbf{n}^{(\kappa)} \in \text{range}(B^*), \quad Z^\top \mathbf{n}^{(\kappa)} = 0. \quad (\text{S36})$$

Set

$$M := Z^\top G_{\mathcal{A}_N^*}. \quad (\text{S37})$$

The effective-space LICQ makes M full column rank. To see this, if $M\mathbf{c} = 0$, then $G_{\mathcal{A}_N^*}\mathbf{c} \in \text{range}(B^*)$; full column rank of $[B^* \ G_{\mathcal{A}_N^*}]$ then forces $\mathbf{c} = 0$.

Projecting Eq. (S24) by $Z^\top$ and using Eqs. (S36)–(S37) gives

$$Z^\top \nabla F(\mathbf{y}^{(\kappa)}) + M \boldsymbol{\omega}_{\mathcal{A}_N^*}^{(\kappa)} = 0. \quad (\text{S38})$$

Consequently,

$$\boldsymbol{\omega}_{\mathcal{A}_N^*}^{(\kappa)} = -(M^\top M)^{-1} M^\top Z^\top \nabla F(\mathbf{y}^{(\kappa)}). \quad (\text{S39})$$

Taking $\kappa \rightarrow \infty$ and using $\mathbf{y}^{(\kappa)} \rightarrow \mathbf{y}^*$ gives the unique solution of the projected hard-KKT stationarity equation. By Eq. (S10), that solution is exactly $\boldsymbol{\omega}_{\mathcal{A}_N^*}^h$. Together with the eventual zeros outside $\mathcal{A}_N^*$, this proves convergence of the full network multiplier vector. $\square$

S3.3 Iterated smooth-to-hard limit

Theorem S1 (Limiting consistency of the smooth scarcity prices). *Under the assumptions in Section S1,*

$$\lim_{\kappa \rightarrow \infty} \left(\lim_{\varepsilon \downarrow 0} \mathbf{y}^{(\kappa, \varepsilon)} \right) = \mathbf{y}^*, \quad (\text{S40})$$

and

$$\lim_{\kappa \rightarrow \infty} \left(\lim_{\varepsilon \downarrow 0} \boldsymbol{\omega}^{(\kappa, \varepsilon)} \right) = \boldsymbol{\omega}^h. \quad (\text{S41})$$

Accordingly,

$$\lim_{\kappa \rightarrow \infty} \lim_{\varepsilon \downarrow 0} (A^V)^\top \boldsymbol{\mu}^{(\kappa, \varepsilon)} = (A^V)^\top \boldsymbol{\lambda}^h, \quad (\text{S42})$$

$$\lim_{\kappa \rightarrow \infty} \lim_{\varepsilon \downarrow 0} (A^C)^\top \boldsymbol{\nu}^{(\kappa, \varepsilon)} = (A^C)^\top \boldsymbol{\nu}^h. \quad (\text{S43})$$

Proof. Apply Proposition S1 for each fixed κ , and then apply Proposition S2. The price-component limits follow because A^V and A^C are fixed linear maps. $\square$

The two propositions are model-specific versions of standard smoothing and quadratic exterior-penalty results; see Refs. [1, 2, 3] and the variational treatment in Ref. [4].

S4 Evaluation points and finite-parameter interpretation

Theorem S1 evaluates multiplier estimates at the corresponding centralized penalty minimizers. When the limiting multiplier is positive, it is recovered through the $O(1/\kappa)$ positive constraint displacement of the quadratic-penalty solution. For example, consider

$$\min_{y \in [-1, 1]} \frac{1}{2}(y - 1)^2 \quad \text{subject to} \quad y \leq 0. \quad (\text{S44})$$

The hard solution is $y^* = 0$ with multiplier $\omega^h = 1$, whereas the quadratic-penalty solution satisfies

$$y^{(\kappa)} = \frac{1}{1 + \kappa}, \quad \omega^{(\kappa)} = \frac{\kappa}{1 + \kappa} \longrightarrow 1. \quad (\text{S45})$$

Evaluation at the fixed hard solution instead gives $\kappa[y^*]_+ = 0$, illustrating why the penalty minimizers are the evaluation points in the theorem.

The scaled smoothing bound at a common argument is

$$0 \leq \kappa \rho_i s_\varepsilon(z) - \kappa \rho_i [z]_+ \leq \kappa \rho_i \varepsilon \log 2. \quad (\text{S46})$$

The stated order, $\varepsilon \downarrow 0$ at fixed κ followed by $\kappa \rightarrow \infty$, first removes the smoothing error and then enforces the hard constraints. A joint-limit result would require a suitable scaling condition and its own proof.

At submitted power profiles, including a QNE of the responsibility-aware game, the finite-parameter coefficients provide explicit differentiable scarcity signals. Theorem S1 concerns centralized OPF minimizers, so it does not identify these coefficients as exact hard-OPF dual variables at an arbitrary profile. Numerical Study I quantifies finite-parameter agreement between the centralized softplus and hard-OPF benchmarks.

S5 Source-Attribution Marginal Feedback and Monotonicity

This section concerns the responsibility-settlement VI at fixed finite penalty parameters and $\varepsilon > 0$. Let $\mathbf{p}^0$ be the fixed reference net-power vector, $\Delta\mathbf{p} := \mathbf{p} - \mathbf{p}^0$, and $\mathbf{p}(\tau) := \mathbf{p}^0 + \tau\Delta\mathbf{p}$. The main-study network maps are $\mathbf{v} = v_0\mathbf{1} - A^V\mathbf{p}$ and $\mathbf{f} = A^C\mathbf{p}$; the corresponding violations are $z_m^V = \underline{v}_m - v_m$ and $z_\ell^C = f_\ell - \bar{f}_\ell$. The row sets $\mathcal{R}_V$ and $\mathcal{R}_C$ contain the constrained buses and monitored lines, respectively. For $X \in \{V, C\}$, the nodal price and its source-attributed change are

$$\begin{aligned}\pi_k^X(\mathbf{p}) &= \sum_{r \in \mathcal{R}_X} A_{rk}^X \kappa_X s_\varepsilon(z_r^X(\mathbf{p})), \\ \pi_{m \leftarrow k}^{X,B} &= \Delta p_k \int_0^1 \frac{\partial \pi_m^X(\mathbf{p}(\tau))}{\partial p_k} d\tau.\end{aligned}$$

The feedback added to the core own-decision marginal is

$$\Delta_k^{\text{attr}} = \sum_{X \in \{V, C\}} \left[\Delta p_k \frac{\partial \pi_k^X}{\partial p_k} + \sum_m p_m^0 \frac{\partial \pi_{m \leftarrow k}^{X,B}}{\partial p_k} \right].$$

The full-space VI operator F_p collects the responsibility-bill marginals minus marginal utility, and $F_y(\mathbf{y}) = G_a^\top F_p(\mathbf{p}^{\text{fix}} + G_a \mathbf{y})$ acts on $\mathcal{Y}^{\text{eff}} = \mathcal{X}$. Here $D_{\text{util}} = \text{diag}(\beta_{u,k})$ is the full nodal-space utility curvature; its effective-space restriction is $D_{\text{util},a} = G_a^\top D_{\text{util}} G_a$, as used in Eq. (S4). The symmetric matrix H^L is the loss matrix in the main manuscript, π^E is the nonnegative wholesale energy price, and v_0 is the fixed root squared voltage.

S5.1 Row feedback and path moments

For $X \in \{V, C\}$ and constraint row $r \in \mathcal{R}_X$, set $a_{rk} := A_{rk}^X$ and $c_r := \sum_m a_{rm} p_m^0$. Throughout this section, s_ε is the softplus function. Along the straight path, define

$$\begin{aligned}z_r^0 &:= z_r^X(\mathbf{p}^0), & b_r &:= \sum_k a_{rk} \Delta p_k, \\ z_r(\tau) &:= z_r^0 + \tau b_r, & z_r^1 &:= z_r(1), \\ t_r &:= \kappa_X s'_\varepsilon(z_r^1), & u_r &:= \kappa_X s''_\varepsilon(z_r^1).\end{aligned}\tag{S47}$$

The continuous path moments are

$$\begin{aligned}T_r &:= \int_0^1 \kappa_X s'_\varepsilon(z_r(\tau)) d\tau, \\ M_r &:= \int_0^1 \tau \kappa_X s''_\varepsilon(z_r(\tau)) d\tau, \\ N_r &:= \int_0^1 \tau^2 \kappa_X s'''_\varepsilon(z_r(\tau)) d\tau.\end{aligned}\tag{S48}$$

Differentiating the source-attributed price path integrals gives the row feedback

$$\Delta_{k,r}^{\text{attr}} = \Delta p_k a_{rk}^2 t_r + c_r (a_{rk} T_r + \Delta p_k a_{rk}^2 M_r).\tag{S49}$$

Differentiating with respect to p_j yields

$$\begin{aligned}[J_{\Delta,r}]_{kj} &= \delta_{kj} a_{rk}^2 t_r + \Delta p_k a_{rk}^2 a_{rj} u_r \\ &\quad + c_r [(a_{rk} a_{rj} + \delta_{kj} a_{rk}^2) M_r \\ &\quad \quad + \Delta p_k a_{rk}^2 a_{rj} N_r],\end{aligned}\tag{S50}$$

where δ_{kj} is the Kronecker delta. Summing over constraint rows and adding utility, loss, and direct voltage/congestion price derivatives gives the complete Jacobian. The direct contribution for channel X is $(A^X)^\top \text{diag}(t_r)A^X$, and the loss contribution is $(2\pi^E/v_0)[H^L + \text{diag}(H_{kk}^L)]$.

For $b_r \neq 0$, integration by parts gives

$$\begin{aligned} T_r &= \frac{\kappa_X[s_\varepsilon(z_r^1) - s_\varepsilon(z_r^0)]}{b_r}, \\ M_r &= \frac{t_r - T_r}{b_r}, \quad N_r = \frac{u_r - 2M_r}{b_r}. \end{aligned} \quad (\text{S51})$$

At zero slope, the exact limits are

$$\begin{aligned} T_r &= \kappa_X s'_\varepsilon(z_r^0), \\ M_r &= \frac{1}{2} \kappa_X s''_\varepsilon(z_r^0), \\ N_r &= \frac{1}{3} \kappa_X s'''_\varepsilon(z_r^0). \end{aligned} \quad (\text{S52})$$

For Gauss–Legendre nodes and weights (τ_q, w_q) on $[0, 1]$, the fixed- Q counterparts are

$$\begin{aligned} T_r^{(Q)} &:= \sum_{q=1}^Q w_q \kappa_X s'_\varepsilon(z_r(\tau_q)), \\ M_r^{(Q)} &:= \sum_{q=1}^Q w_q \tau_q \kappa_X s''_\varepsilon(z_r(\tau_q)), \\ N_r^{(Q)} &:= \sum_{q=1}^Q w_q \tau_q^2 \kappa_X s'''_\varepsilon(z_r(\tau_q)). \end{aligned} \quad (\text{S53})$$

The solver uses common nodes and weights for the bills, path moments, and operator $F_y^{(Q)}$. Replacing (T_r, M_r, N_r) by their fixed- Q counterparts in the row formulas gives the corresponding discrete feedback and Jacobian; the endpoint terms t_r and u_r are unchanged.

The continuous-curvature diagnostics reported for the demand model in Numerical Studies V–VI instead use Eqs. (S51) and (S52). A fast evaluation first scans the prescribed points. The selected extrema are then recomputed with 80-digit analytic evaluation and checked independently by segmented 96-point Gauss–Legendre integration. The normalized Jacobian and minimum-eigenvalue discrepancies must not exceed 10^{-10} and 10^{-8} , respectively; otherwise the curvature sign is recorded as unresolved. This check concerns the continuous Jacobian. The separate endpoint validation compares $F_y^{(Q)}$ with breakpoint-assisted adaptive integration, as described in the main manuscript’s computational workflow.

S5.2 Symmetric Jacobian and global certificate

Let $S_p := (\nabla F_p + \nabla F_p^\top)/2$ and define

$$\begin{aligned} S_{\text{loss}} &:= \frac{2\pi^E}{v_0} [H^L + \text{diag}(H_{kk}^L)], \\ S_X^{\text{dir}} &:= (A^X)^\top \text{diag}(t_r)A^X, \\ S_y^{\text{smooth}} &:= G_a^\top [D_{\text{util}} + S_{\text{loss}} + S_V^{\text{dir}} + S_C^{\text{dir}}]G_a. \end{aligned}$$

In the effective decision space, the symmetric Jacobian is

$$\begin{aligned} S_y &= G_a^\top S_p G_a = S_y^{\text{smooth}} + S_{\Delta \text{attr}, y}, \\ S_{\Delta \text{attr}, y} &= G_a^\top \frac{J_\Delta + J_\Delta^\top}{2} G_a, \quad J_\Delta = \sum_{X, r} J_{\Delta, r}. \end{aligned} \quad (\text{S54})$$

The core contribution contains positive utility curvature and nonnegative loss and direct-penalty curvature. Equation (S50) shows how source deviations, sensitivities, and the first three soft-plus derivatives determine the remaining symmetric contribution. This contribution can offset core curvature; its eigenvalues, rather than Jacobian asymmetry alone, determine its effect on monotonicity.

A uniform bound $S_y \succeq \sigma I$ with $\sigma > 0$ certifies QNE uniqueness by the main manuscript's QNE uniqueness theorem. To restate its global certificate, let $J_{\Delta,y} := G_a^\top J_\Delta G_a$ and choose verified bounds

$$\begin{aligned} m_{\text{core}}^{\text{LB}} &\leq \inf_{\mathbf{y} \in \mathcal{Y}^{\text{eff}}} \lambda_{\min}(S_y^{\text{smooth}}(\mathbf{y})), \\ |J_{\Delta,y}(\mathbf{y})| &\leq M_\Delta \quad \text{elementwise on } \mathcal{Y}^{\text{eff}}, \quad M_\Delta \geq 0, \\ B_{\Delta\text{attr}}^{\text{ub}} &:= \sqrt{\|M_\Delta\|_1 \|M_\Delta\|_\infty}, \\ \sigma_{\text{LB}}^{\text{glob}} &:= m_{\text{core}}^{\text{LB}} - B_{\Delta\text{attr}}^{\text{ub}}. \end{aligned} \tag{S55}$$

The matrix-norm bound and Weyl's inequality give $\lambda_{\min}(S_y(\mathbf{y})) \geq \sigma_{\text{LB}}^{\text{glob}}$ throughout the feasible box, so a positive bound certifies strong monotonicity. Continuous-operator evaluations instead characterize curvature at the tested operating points.

S5.3 Construction of the verified bounds

Let $[\underline{\mathbf{p}}, \bar{\mathbf{p}}]$ be the reachable full-power box. Fixed physical coordinates have equal endpoints in this box. Every attribution path from $\mathbf{p}^0$ to a reachable profile lies in the enclosing box

$$\underline{\mathbf{p}}^{\text{path}} := \min(\underline{\mathbf{p}}, \mathbf{p}^0), \quad \bar{\mathbf{p}}^{\text{path}} := \max(\bar{\mathbf{p}}, \mathbf{p}^0), \tag{S56}$$

where minimum and maximum are elementwise. This construction also covers a settlement baseline outside the physical decision box, including settlement deviations at fixed-load buses.

Write a row violation as $z_r^X(\mathbf{p}) = \eta_r + \sum_k a_{rk} p_k$. Its affine extrema over the path box are

$$\begin{aligned} z_r^- &:= \eta_r + \sum_k \min\{a_{rk} \underline{p}_k^{\text{path}}, a_{rk} \bar{p}_k^{\text{path}}\}, \\ z_r^+ &:= \eta_r + \sum_k \max\{a_{rk} \underline{p}_k^{\text{path}}, a_{rk} \bar{p}_k^{\text{path}}\}. \end{aligned} \tag{S57}$$

For softplus, verified upper bounds for the first three derivatives can be obtained from

$$\begin{aligned} D_{1,r} &:= s'_\varepsilon(z_r^+), \\ D_{2,r} &:= s''_\varepsilon\left(\Pi_{[z_r^-, z_r^+]}(0)\right), \\ D_{3,r} &:= \max_{z \in \mathcal{K}_r} |s'''_\varepsilon(z)|, \end{aligned} \tag{S58}$$

where $\mathcal{K}_r$ contains the two interval endpoints and any of $\pm\varepsilon \log(2 + \sqrt{3})$ lying in $[z_r^-, z_r^+]$. The first derivative is increasing, the second is largest at the point nearest zero, and the absolute third derivative has its global peaks at these two interior candidates, with value $\sqrt{3}/(18\varepsilon^2)$. In verified computation, Eq. (S58) is evaluated with conservative upper endpoints of outward-rounded intervals.

Define

$$d_k := \max\{|\underline{p}_k - p_k^0|, |\bar{p}_k - p_k^0|\}, \quad \bar{c}_r := \sum_m |a_{rm} p_m^0| \geq |c_r|. \tag{S59}$$

The path-moment definitions give $|M_r| \leq \kappa_X D_{2,r}/2$ and $|N_r| \leq \kappa_X D_{3,r}/3$, while $t_r \leq \kappa_X D_{1,r}$

and $|u_r| \leq \kappa_X D_{2,r}$. Taking absolute values term by term in Eq. (S50) therefore yields

$$\begin{aligned} [M_{\Delta,r}]_{kj} &:= \kappa_X [\delta_{kj} |a_{rk}|^2 D_{1,r} + d_k |a_{rk}|^2 |a_{rj}| D_{2,r} \\ &\quad + \frac{\bar{c}_r}{2} (|a_{rk} a_{rj}| + \delta_{kj} |a_{rk}|^2) D_{2,r} \\ &\quad + \frac{\bar{c}_r d_k}{3} |a_{rk}|^2 |a_{rj}| D_{3,r}]. \end{aligned} \quad (\text{S60})$$

For the demand-only embedding, G_a selects the flexible coordinates. Summing these nonnegative row bounds over voltage and congestion rows and restricting to those coordinates gives

$$M_{\Delta} = G_a^{\top} \left(\sum_{X,r} M_{\Delta,r} \right) G_a, \quad |J_{\Delta,y}(\mathbf{y})| \leq M_{\Delta}. \quad (\text{S61})$$

Let $L_a := G_a^{\top} S_{\text{loss}} G_a$. The implementation uses the following core lower bound:

$$\begin{aligned} m_{\text{core}}^{\text{LB}} &:= \min_h \beta_{u,h} + \ell_{\text{loss}}^{\text{LB}}, \\ \ell_{\text{loss}}^{\text{LB}} &:= \min_h \left[(L_a)_{hh} - \sum_{j \neq h} |(L_a)_{hj}| \right], \end{aligned} \quad (\text{S62})$$

where h indexes the retained flexible coordinates. Gershgorin's bound controls the minimum eigenvalue of the symmetric loss matrix, and the positive-semidefinite direct voltage and congestion contributions are discarded conservatively. This proves the required core inequality in Eq. (S55). Combining it with Eq. (S61) gives the stated global certificate. A nonpositive $\sigma_{\text{LB}}^{\text{glob}}$ means that this sufficient condition has not been established; it does not establish multiple QNEs.

The certificate implementation treats each supplied IEEE-754 coefficient, including precomputed violation offsets and the loss scaling, as an exact input to 160-bit interval arithmetic with outward rounding. Lower and upper bounds are also rounded outward when converted to floating-point outputs. The certificate concerns this fixed input model; it does not enclose uncertainty in the original data or upstream matrix construction.

S5.4 Signed demand and generation actions

For the signed extension in the manuscript's Appendix A, the same network row formulas apply after including lower- and upper-voltage limits and forward- and reverse-flow limits. Their row gradients are respectively $+A^V$, $-A^V$, $+A^C$, and $-A^C$, with the appropriate affine offsets. The baseline weight $c_r = \sum_m a_{rm} p_m^0$ may be negative; the absolute bound $\bar{c}_r$ is used only for certification.

Let $\mathbf{z}$ collect the flexible demand and generation actions, with original plans $\hat{\mathbf{z}}$. Their net-power embedding is

$$\mathbf{p}(\mathbf{z}) = \mathbf{p}^{\text{plan}} + E_z(\mathbf{z} - \hat{\mathbf{z}}), \quad (\text{S63})$$

where a demand column of E_z is $+\mathbf{e}_k$ and a generation column is $-\mathbf{e}_k$. The physical net-power plan $\mathbf{p}^{\text{plan}} = \hat{\mathbf{d}} - \hat{\mathbf{g}}$ is distinct from the settlement baseline $\mathbf{p}^0$. Writing $\boldsymbol{\xi}(\mathbf{p})$ for the responsibility-bill marginals in net-power space, and $\mathbf{f}_{\text{role}}$ for the negative marginal demand utilities and positive marginal generation costs, gives

$$\begin{aligned} F_z(\mathbf{z}) &= \mathbf{f}_{\text{role}}(\mathbf{z}) + E_z^{\top} \boldsymbol{\xi}(\mathbf{p}(\mathbf{z})), \\ J_z(\mathbf{z}) &= D_{\text{role}} + E_z^{\top} J_{\boldsymbol{\xi},p} E_z, \quad J_{\boldsymbol{\xi},p} := \nabla_{\mathbf{p}} \boldsymbol{\xi}. \end{aligned} \quad (\text{S64})$$

Here D_{role} is diagonal, with demand entries $-U_k'' = a_{d,k} > 0$ and generation entries $C_k'' = a_{g,k} > 0$ for the quadratic role models. Thus $J_{\xi,p}$ includes bill curvature only; utility and generation-cost curvature are added directly in action space. The symmetric action Jacobian is

$$S_z = D_{\text{role}} + E_z^\top \frac{J_{\xi,p} + J_{\xi,p}^\top}{2} E_z. \quad (\text{S65})$$

These equations extend the continuous mathematical formulas to signed actions. The signed implementation's stored local minimum-eigenvalue diagnostics instead use the native fixed- Q Jacobian obtained from Eq. (S53). They have not undergone the continuous-curvature validation described above for Studies V–VI. Likewise, the demand-space certificate construction above does not by itself certify the signed cases: an action-space certificate must bound the embedded bill-feedback Jacobian and use D_{role} in its core lower bound.

References

- [1] C. Chen and O. L. Mangasarian, “A class of smoothing functions for nonlinear and mixed complementarity problems,” *Computational Optimization and Applications*, vol. 5, pp. 97–138, 1996.
- [2] A. V. Fiacco and G. P. McCormick, *Nonlinear Programming: Sequential Unconstrained Minimization Techniques*. Wiley, 1968.
- [3] D. P. Bertsekas, *Nonlinear Programming*, 2nd ed. Athena Scientific, 1999.
- [4] R. T. Rockafellar and R. J.-B. Wets, *Variational Analysis*. Springer, 1998.